

Samuel James Boyer
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Forward Deployed Engineer - Neural Engineering and BCI specializing in real-time low-latency bioelectric pipelines and field-deployed AI workflows.

Education

Case Western Reserve University | Cleveland, OH

M.S. in Biomedical and Health Informatics (Part-Time) Expected May 2028

- Concentration: Artificial Intelligence for Neural Engineering
- Graduate Coursework: AI in Medicine and Data Science Systems

B.S.E. in Biomedical Engineering (Minor in Electrical Engineering) May 2024

Experience

ReHAB (Reconnecting Hand-Arm-Brain) Clinical Trial | Cleveland, OH

Software Engineer I July 2024 – Present

- Agentially redesigned our C++ neural processing pipeline using lock-free multi-threading.
- Achieved a 12x development acceleration using an agentic graph workflow, parallelizing deployment of powerful models for design, implementation, and review.
- Implemented flexible harnesses with LangGraph, allowing agents to dynamic rebalance across the cloud and review their own code.
- Designed neural network models for regression and classification analysis of neural dynamics and disease states with PyTorch.
- De-risked development using docker containerization and local LLMs to maintain strict separation between autonomous agents and sensitive data.
- Engineered agentic feedback loops with medical-grade neurostimulators, allowing output waveforms to feedback into the agent's environment, allowing continuous improvement, safety checks, and debugging in high-stakes medical risk.
- Developed internal latency profiling and benchmark instruments to automate system debugging and telemetry tracking across days.
- Built researcher-facing control UI in C#/Python for experiment deployment and tuning.
- Spearheaded lab-wide AI integration strategy, co-authoring 2 AI-focused grant proposals for advanced neural decoding research.

- Owned end-to-end experimental setup and execution for live human clinical trials, performing real-time hardware/software debugging under strict clinical constraints.
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Technical Skills

- Programming & Frameworks: C/C++, Python, C#, MATLAB, Docker, SLURM, Github
- Front-end Platforms: Meta Quest SDK, Unity Editor, Dear PyGUI
- Data Systems: Redis, Supabase
- AI & Systems Architecture: MCP Development with Fast API, Cloud/Sandbox Agent Deployment, Skills Engineering, Cursor IDE